Radio Emission at 144 MHz and 3 GHz from the T8.5+T9.5 Binary WISE 0458+6434

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Abstract

Electron-cyclotron maser emission from brown dwarfs measures the magnetic fields of objects too cold for any other magnetic diagnostic, and low frequencies reach fields of tens of gauss. The third data release of the LOFAR Two-metre Sky Survey (LoTSS-DR3) holds a component of 1.17 ± 0.18 mJy at 144 MHz 0.88″ from the T8.5+T9.5 binary WISEPC J045853.89+643452.9 (WISE 0458+6434) at 9.4 pc, where a component that bright falls that close to any of the 64 T and Y dwarfs DR3 covers with probability 0.003. In DR3's per-pointing images of five epochs from 2022 December to 2023 August the circular polarisation at the binary's position is -0.395 ± 0.085 mJy beam^{-1} (-4.6σ), about 42 per cent of the total intensity and the same within the errors at every epoch, where 16 compact sources of the same field read a fractional circular polarisation of +0.003 and +0.011. The quick-look images of the VLA Sky Survey hold a 3 GHz source of 472–634 μJy beam^{-1} (3.9–5.4σ) within 0.54″ of the binary's position on each of 2017 October 11, 2023 February 9 and 2025 September 6, across the 2.8″ the binary moved, and nothing on 2020 August 25, where a source as bright would stand at 3.4σ or more; the three positions fit the binary's track ($\chi^2 = 1.3$ for 6 degrees of freedom) and reject a fixed source ($\chi^2 = 16.1$ for 4, $p = 0.003$). The spectral luminosities, 1.2×10^{14} erg s^{-1} Hz^{-1} at 144 MHz and 5×10^{13} to 7×10^{13} erg s^{-1} Hz^{-1} at 3 GHz, the strong circular polarisation and the brightness temperature are those of auroral electron-cyclotron emission, which at 3 GHz requires a field of about 1.1 kG where it forms. Both components are later than the T8 dwarf WISE J062309.94−045624.6, the coldest brown dwarf detected at radio wavelengths before.

1 Introduction

Ultracool dwarfs, from the bottom of the main sequence down through the brown dwarfs, carry magnetic fields of kilogauss strength, which they show through radio emission: electrons accelerated along the field lines of a large-scale magnetosphere radiate by the electron-cyclotron maser instability at the local electron gyrofrequency, $\nu = 2.8\,B$ MHz for B in gauss, producing bright, strongly circularly polarised and often periodic emission, the radio counterpart of the aurorae that the same currents drive [1, 2]. At 4–8 GHz this emission has been found from late L and T dwarfs with fields above 2.5 kG [3], and Arecibo saw sporadic polarised flares from the T6.5 dwarf 2MASS J1047+21 [4]. The coldest brown dwarf detected at radio wavelengths is the T8 dwarf WISE J062309.94−045624.6, whose periodic emission the Australian Square Kilometre Array Pathfinder found [5]. Three Y dwarfs observed at 4–8 GHz with the Jansky VLA gave no detection [6], and a search of the VLA Sky Survey (VLASS) catalogues and images at 14915 ultracool dwarf candidates of types M7.5 to L4 found no counterpart [7].

Frequencies near 144 MHz probe fields near 50 G, which a gigahertz search cannot see. The LOFAR Two-metre Sky Survey (LoTSS) found the T6.5 dwarf BDR J1750+3809 from its circularly polarised emission before it was known as a brown dwarf [8], and the T5.5+T7 binary WISEP J101905.63+652954.2 the same way, pulsing with a period of about three hours [9], with a field of

about 126 G at its poles from the cutoff of its radio spectrum [10]. The circularly polarised catalogue of the second data release held no other ultracool dwarf [11]. The third data release (DR3) extends the survey over the northern sky [12], and this note reports its detection of a T dwarf later than any of these, together with that source's circular polarisation in DR3's per-pointing images and its 3 GHz counterpart in the VLASS quick-look images.

1.1 Contributions

1. A search of LoTSS-DR3 at the T and Y dwarfs of the 20 pc census north of the equator, which finds one, WISE 0458+6434.
2. The source's Stokes V in the per-pointing images of the five DR3 epochs covering it, read from the survey's archives by byte range, against the field's compact sources and random positions.
3. Its 3 GHz counterpart in three of the four VLASS epochs, moving with the binary.

2 WISE 0458+6434

WISEPC J045853.89+643452.9 was the first ultracool brown dwarf the Wide-field Infrared Survey Explorer found [13]. Keck laser guide star adaptive optics resolved it into two components 0.51″ apart at position angle 320°, with estimated temperatures near 600 and 500 K [14], and resolved spectroscopy classifies them T8.5 and T9.5, at the boundary between the T and Y classes, with ammonia absorption in the secondary [15]. The 20 pc census gives a parallax of 106.7 mas, a distance of 9.37 pc and so a projected separation of 4.8 au, and a proper motion of 210.4 mas yr^{-1} east and 289.6 mas yr^{-1} north, 358 mas yr^{-1} at position angle 36° [16]. Positions in this note are carried from the census's position at epoch 2016.45 along that motion.

3 Data and method

The DR3 component catalogue. The 156 objects of the Gaia Catalogue of Nearby Stars within 10 pc and the 80 T and Y dwarfs of the 20 pc census north of the equator were carried to 2019.5 and matched to the DR3 component catalogue (VizieR J/A+A/707/A198) within 3″, the same positions moved 2′ in four directions serving as the null. The chance of a match follows from DR3's local density of components.

The per-pointing images. The DR3 mosaic at a position combines several pointings, each an 8 h observation of one epoch. The ASTRON VO's table of DR3 pointings gives their dates, and the SURF Data Repository holds each pointing's products as tar archives of 10–23 GB, from which the 6″ Stokes I image with astrometric corrections and the 6″ dirty Stokes I and V images were read by HTTP byte range, a patch around the target alone. Each pointing's images are put on DR3's flux scale by the factor the release gives for that pointing. Stokes V is read at the binary's position on the pointing's date, its noise from the robust scatter of the annulus 12″–30″ out, and epochs are combined by inverse variance. The field's other compact sources and random positions in the same images are the null.

VLASS. The quick-look images of the four VLASS epochs [17] come from the Canadian Astronomy Data Centre as cutouts. In each, the brightest pixel within 2.5″ of the binary's position on the image's date is set against the image's robust noise, a Gaussian fitted to it gives its offset, and the

same statistic at 2000 random positions 15″–150″ away gives the rate at which noise alone reaches it. Quick-look positions carry systematic errors of about 0.5″ [18], which are added in quadrature to the fitted errors.

4 Results

4.1 The 144 MHz source

Of the 126 stars within 10 pc inside DR3's coverage, 7 carry a component within 3″, against 0.25 at the moved positions: the flare stars YZ CMi, AD Leo, EV Lac, EQ Peg, V374 Peg and the pair GJ 1116 A and B. Of the 64 T and Y dwarfs inside the coverage, one does, against 0.00 moved. WISE 0458+6434 lies under a component of 1.17 ± 0.18 mJy with a peak of 0.90 mJy beam^{-1} (signal-to-noise ratio 11), unresolved at $7.6'' \times 6.1''$, 0.88″ from the binary's 2019.5 position at position angle 45°, against the proper motion's 36°: the DR3 epochs fall in 2022 and 2023, and the binary had moved 1.2″ to 1.5″ along that direction by then. DR3's local density of 275 components above 1 mJy per square degree puts a component that bright within 0.88″ of any of the 64 dwarfs with probability 0.003. Nothing lies within 10″ in NVSS, the VLASS epoch-1 component catalogue, WENSS, Pan-STARRS1 or Gaia, and AllWISE's only source there is the binary.

4.2 Circular polarisation

Table 1 gives the five pointings whose images cover the binary. In the two nearest the pointing centre, P078+64 of 2022 December 29 and P072+64 of 2023 July 14, the restored Stokes I image holds the source at signal-to-noise ratios of 10.8 and 4.6, 0.36″ and 0.98″ from the binary's position on each date, with Stokes V at the Stokes I peak of -0.383 ± 0.118 and -0.253 ± 0.116 mJy beam^{-1} before the flux scale, fractions of -0.44 ± 0.14 and -0.43 ± 0.23 of the dirty Stokes I there. The 16 other compact sources of 1.1–6.8 mJy beam^{-1} within 10.5′ give median fractions of +0.003 and +0.011 in the two epochs, none beyond 2.7σ; no pixel of the two epochs' combined Stokes V map more than 30″ from the binary reaches 4.7σ of either sign among about 33800 beams; and at 195419 random positions both epochs reach the binary's values with one sign at a rate of 2×10^{-5}.

All five epochs on DR3's flux scale give -0.395 ± 0.085 mJy beam^{-1} at the binary's position (-4.6σ), with $\chi^2 = 2.4$ for 4 degrees of freedom about that value, and Stokes I of 0.93 ± 0.11 mJy beam^{-1}, a fractional polarisation of about −0.42. A reading of the same images through the two nearest pointings, part of a survey of the nearest stars, gives -4.1σ. Among the 1631 objects SIMBAD lists as brown dwarfs north of −1° that DR3's per-pointing images measure, two exceed 3.5σ in Stokes V: WISE 0458+6434 and WISEP J101905.63+652954.2 (-3.8σ), the binary LOFAR found before [9]. The Stokes I peak of the two deepest epochs differs by a factor of 0.46 ± 0.11 against 0.83 ± 0.14 for the field's sources, a difference of 2σ.

The visibilities of the 2023 July 14 observation (25 sub-bands, 56.6 GB) were read from the archive and phase-rotated to the binary, to four positions 45″ away and to a 7.78 mJy source of the field. Without the archive's direction-dependent calibration solutions, the check source keeps about a tenth of its flux, and the binary's Stokes V series binned from 8 s to 30 min holds no excursion beyond 4σ of the offset positions' scatter and no period between 0.5 and 4 h (false-alarm probability 0.60). At that calibration a burst would have to reach of order 100 mJy for a minute to show.

Table 1: WISE 0458+6434 in the per-pointing images of LoTSS-DR3: the pointing, its date, the binary's distance from the pointing centre, DR3's flux-scale factor, the dirty Stokes I peak within 4.5″ of the binary and Stokes V at the binary's position, both scaled. The low-S/N peaks are biased upward by the search for a maximum.

Pointing	Date	Offset (°)	Scale	I (mJy beam^{-1})	V (mJy beam^{-1})
P072+67	2022 Dec 28	2.34	1.159	0.60 ± 0.31	-0.13 ± 0.25
P078+64	2022 Dec 29	1.62	1.052	1.25 ± 0.19	-0.40 ± 0.13
P072+64	2023 Jul 14	1.09	1.181	0.74 ± 0.16	-0.39 ± 0.14
P075+62	2023 Aug 19	2.93	0.991	1.32 ± 0.55	-0.71 ± 0.53
P078+67	2023 Aug 20	2.66	1.132	1.00 ± 0.39	-0.69 ± 0.32
Combined				0.93 ± 0.11	-0.395 ± 0.085

Table 2: WISE 0458+6434 in the VLASS quick-look images: the epoch, the date, the beam, the brightest pixel within 2.5″ of the binary's position on the date and its signal-to-noise ratio, the fitted position's offset east and north of the binary, and the share of 2000 random positions where the same statistic is as high.

Epoch	Date	Beam (″)	Peak (μJy beam^{-1})	S/N	Offset E, N (″)	Random
1.1	2017 Oct 11	3.1×2.1	634	5.4	+0.07, −0.32	0.0000
2.1	2020 Aug 25	3.3×2.1	13	0.1	–	–
3.1	2023 Feb 9	2.6×2.2	477	4.3	+0.00, −0.23	0.0000
4.1	2025 Sep 6	2.6×2.0	472	3.9	+0.54, −0.02	0.0005

4.3 The 3 GHz source

Table 2 and Figure 1 give the VLASS epochs. In 2017, 2023 and 2025 the brightest pixel within 2.5″ of the binary stands at 3.9–5.4σ, 0.23″–0.54″ from the binary's position on the date, a level that 2000 random positions in the same images reach once in the three epochs; in 2020 the binary's position holds nothing (0.1σ, a 5σ limit of 0.7 mJy). Over the eight years the binary moved 2.8″. The three positions, with the fitted and systematic errors in quadrature, give $\chi^2 = 1.3$ for 6 degrees of freedom about the binary's track and $\chi^2 = 16.1$ for 4 about the best fixed position ($p = 0.003$), so the source moves with the binary; the two LOFAR positions of 2022–2023 lie on the track as well, the five together giving $\chi^2 = 12.1$ for 10 about the track and 18.8 for 8 about a fixed position.

5 Discussion

At 9.37 pc the 144 MHz flux density corresponds to a spectral luminosity of 1.2×10^{14} erg s^{-1} Hz^{-1}, of which 4×10^{13} erg s^{-1} Hz^{-1} is circularly polarised, and the VLASS detections to 5.0×10^{13}–6.7×10^{13} erg s^{-1} Hz^{-1} at 3 GHz. Emission from a region of radius R at 144 MHz has a brightness temperature of $1 \times 10^{13}\,(R/R_{\rm J})^{-2}$ K, above the 10^{12} K that incoherent synchrotron emission reaches for any region smaller than about three Jupiter radii, and a circular polarisation of 42 per cent at 144 MHz is not that of gyrosynchrotron emission from an optically thick region. Both point to the electron-cyclotron maser, which emits near the local gyrofrequency: 144 MHz needs about 51 G where it forms and 3 GHz about 1.1 kG, or half these at the second harmonic. A surface field of a kilogauss or more on one of the components, which 4–8 GHz detections have shown in T dwarfs before [3], would supply both frequencies from heights of about three radii and from near

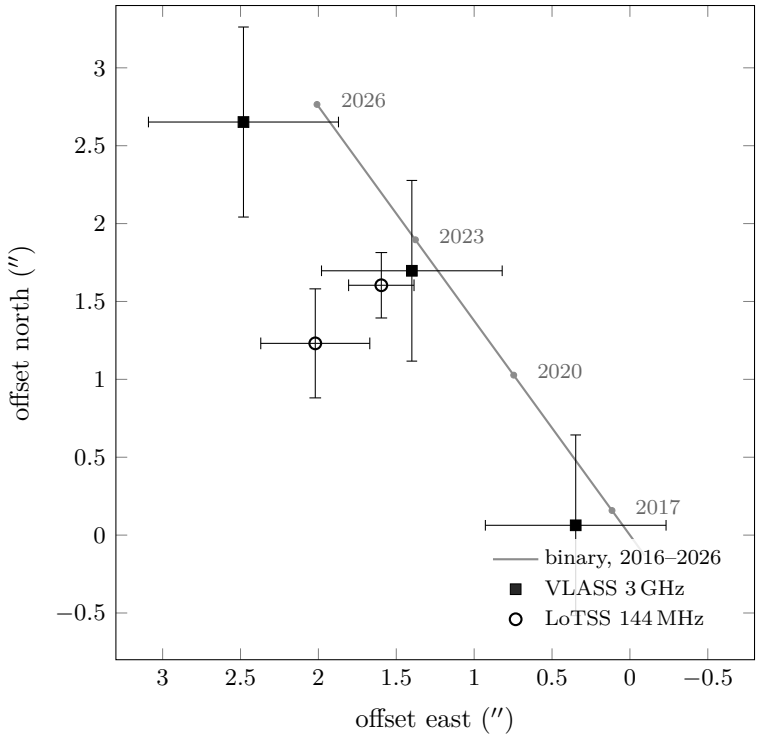


Figure 1: Positions of the radio source relative to WISE 0458+6434's position at epoch 2016.45, east to the left. The line is the binary's track from 2016 to 2026, marked at 2017, 2020, 2023 and 2026; the squares are the VLASS detections of 2017, 2023 and 2025 and the circles the two deepest LoTSS-DR3 epochs, of 2022 December and 2023 July, each with its fitted and systematic errors in quadrature.

the surface. In 2020 the binary's position reads $13\,\mu\mathrm{Jy\,beam^{-1}}$ against a noise of $140\,\mu\mathrm{Jy\,beam^{-1}}$, where a source as bright as in the other epochs would stand at 3.4–4.5σ, as the emission's beaming across the rotation or its variability from epoch to epoch would make it in the seconds-long VLASS snapshots; the 144 MHz polarisation is the same within its errors over the five 8 h epochs.

The two components lie $0.51''$ apart, unresolved by either survey, so the emitter is not identified. Both are later than the T8 dwarf WISE J062309.94−045624.6, the coldest brown dwarf detected at radio wavelengths before [5], and at T8.5+T9.5 the binary is the latest-type brown dwarf system detected at any radio frequency and the first at the boundary of the T and Y classes. The orbit of about 70 yr [14] is too wide for the components' magnetospheres to interact, so the emission is each dwarf's own.

The chance that an unrelated source accounts for the detection is small on each count: 0.003 for a 144 MHz component that close among the 64 dwarfs, 2×10^{-5} for Stokes V of the observed strength and sign at a random position in both deep epochs, and 0.003 for the VLASS positions under a fixed source, which also leaves the binary's motion between 2017 and 2025 unexplained. A background active galaxy would carry a circular polarisation below one per cent, as the field's 16 compact sources do.

6 Related work

The radio emission of brown dwarfs has been found mostly by targeted gigahertz observations [4, 3, 5], and at 144 MHz by the LoTSS search for circularly polarised sources [8, 9, 11]. The VLASS search of ultracool dwarfs found none [7]; its sample was the M7.5 to L4 candidates of a catalogue built on Gaia DR2, which holds no T dwarfs, and the VLASS epoch-1 component catalogue does not list the 2017 source here. The VLA observations of three Y dwarfs reached 2–7 μJy rms at 4–8 GHz without a detection [6]. No earlier radio detection of WISE 0458+6434 is found in the literature.

7 Conclusion

WISE 0458+6434, a T8.5+T9.5 binary at 9.4 pc, emits at 144 MHz with a circular polarisation of about 42 per cent over five epochs of 2022–2023 and at 3 GHz in three of four VLASS epochs from 2017 to 2025, the 3 GHz source moving with the binary. The luminosity, polarisation and brightness temperature are those of auroral electron-cyclotron emission, which at 3 GHz requires a field of about a kilogauss where it forms, and the binary is the latest-type brown dwarf system detected at radio wavelengths. Time-resolved observations at either frequency would give the emitter's rotation period through its pulses, the polarisation at 3 GHz would fix the emission's mode, and resolved observations would identify the component.

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